MODEL EQUATIONS · LBM_DROPLET_FIG3.PY

Droplet LBM Equations

The 2D model behind the re-creation of Fig. 3 of Monluc et al., Nano Energy 146 (2025) 111536: a charged water droplet that falls on, slides down and leaves an inclined PTFE/copper plate. Each part shows the normal (continuum) equation first, then the lattice Boltzmann or discrete form the code steps through. A colour means the same physical role everywhere on the page.

Time change
∂/∂t, the new value after a step

Transport by the flow
advection, streaming

Diffusion & viscosity
D, ν, wall friction

Interface & pressure
Shan–Chen, wetting, keeping ions in water

Forces & fields
gravity, electric field, charges

Collision & equilibrium
relaxation toward the equilibrium

What happens in one time step

1 Every 50 steps: solve the Poisson equation for **E** and the electrode charge Q `poisson()`

2 Flow: update the PTFE wall density, collide (MRT + force) `k_collide_mrt`

3 Flow: stream, with bounce-back and Navier slip at walls `k_stream9`

4 Ions: collide and stream (D2Q5) `_ion_step`

1 Water and air flow

NORMAL EQUATION · WEAKLY COMPRESSIBLE NAVIER–STOKES WITH A NON-IDEAL PRESSURE

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

$$\frac{\partial (\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u}\mathbf{u}) = -\nabla p + \nabla \cdot \left[\rho \nu (\nabla \mathbf{u} + \nabla \mathbf{u}^T)\right] + \rho \mathbf{g}$$

$$p = \rho c_s^2 + \frac{G c_s^2}{2} \psi(\rho)^2, \qquad \psi(\rho) = 1 - e^{-\rho}, \qquad c_s^2 = \tfrac{1}{3}, \qquad \mathbf{g} = g\,(\sin\alpha,\ -\cos\alpha)$$

Water and air separate by themselves because of the non-ideal pressure p(ρ); surface tension comes out of it, so there is no separate surface-tension term. The water fraction is $m = (\rho - \rho_v)/(\rho_l - \rho_v)$. The plate is kept horizontal and gravity is tilted by α = 30°.

> For comparison, the paper (COMSOL) solves the incompressible form with a Cahn–Hilliard phase field ϕ:
>
> $$\rho\left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u}\cdot\nabla\mathbf{u}\right) = -\nabla p + \mu\nabla^2\mathbf{u} + \rho\mathbf{g} + \mathbf{F}_e + \mathbf{F}_s, \quad \nabla\cdot\mathbf{u} = 0, \quad \frac{\partial \phi}{\partial t} + \mathbf{u}\cdot\nabla\phi =$$

discretised as

LATTICE BOLTZMANN · D2Q9 MULTIPLE-RELAXATION-TIME (MRT), EXACT-DIFFERENCE FORCING

k_collide_mrt · k_stream9

$$f_i(\mathbf{x} + \mathbf{e}_i,\ t+1) = f_i(\mathbf{x}, t) \ - \ \left[\mathbf{M}^{-1}\,\mathbf{S}\,\left(\mathbf{m} - \mathbf{m}^{eq}\right)\right]_i \ + \ \Delta F_i \qquad i = 0, \ldots, 8$$

$$f_i^{eq}(\rho, \mathbf{u}) = w_i\,\rho\left[1 + 3\,\mathbf{e}_i\cdot\mathbf{u} + \tfrac{9}{2}(\mathbf{e}_i\cdot\mathbf{u})^2 - \tfrac{3}{2}\,u^2\right]$$

$$\Delta F_i = f_i^{eq}\left(\rho,\ \mathbf{u}_0 + \frac{\mathbf{F}}{\rho}\right) - f_i^{eq}(\rho, \mathbf{u}_0) \qquad \text{(exact-difference method)}$$

$$\mathbf{F}(\mathbf{x}) = -\,G\,\psi(\mathbf{x})\sum_i w_i\,\psi(\mathbf{x} + \mathbf{e}_i)\,\mathbf{e}_i \ + \ \rho\,\mathbf{g} \qquad \text{(Shan–Chen force + gravity)}$$

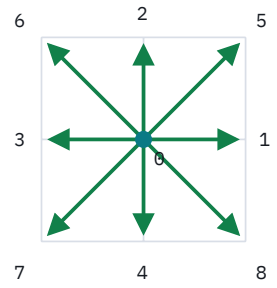
$$\rho = \sum_i f_i, \qquad \rho\,\mathbf{u}_0 = \sum_i f_i\,\mathbf{e}_i, \qquad \mathbf{u} = \mathbf{u}_0 + \frac{\mathbf{F}}{2\rho}, \qquad \nu = \tfrac{1}{3}\left(\tau - \tfrac{1}{2}\right)$$

$$\mathbf{m} = \mathbf{M}\mathbf{f}, \qquad \mathbf{M} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -4 & -1 & -1 & -1 & -1 & 2 & 2 & 2 & 2 \\ 4 & -2 & -2 & -2 & -2 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & -1 & 0 & 1 & -1 & -1 & 1 \\ 0 & -2 & 0 & 2 & 0 & 1 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & -1 & 1 & 1 & -1 & -1 \\ 0 & 0 & -2 & 0 & 2 & 1 & 1 & -1 & -1 \\ 0 & 1 & -1 & 1 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 & -1 \end{pmatrix} \begin{matrix} \rho \\ e \\ \varepsilon \\ j_x \\ q_x \\ j_y \\ q_y \\ p_{xx} \\ p_{xy} \end{matrix}$$

$$\mathbf{m}^{eq} = \left(\rho,\ \rho(-2 + 3u^2),\ \rho(1 - 3u^2),\ \rho u_x,\ -\rho u_x,\ \rho u_y,\ -\rho u_y,\ \rho(u_x^2 - u_y^2),\ \rho u_x u_y\right)^T$$

$$\mathbf{S} = \mathrm{diag}\left(0,\ s_e,\ s_\varepsilon,\ 0,\ s_q,\ 0,\ s_q,\ \tfrac{1}{\tau},\ \tfrac{1}{\tau}\right)$$

τ = 0.53 (ν = 0.01) · s_e = s_ε = 1.0 · s_q = 1.1 · G = −5 · ρ_l = 1.952 · ρ_v = 0



D2Q9: $w_0 = 4/9$, $w_{1\ldots4} = 1/9$, $w_{5\ldots8} = 1/36$

If all rates in **S** were 1/τ, the update would reduce to the plain single-relaxation (BGK) equation $f_i(\mathbf{x}+\mathbf{e}_i, t+1) = f_i - \frac{1}{\tau}(f_i - f_i^{eq}) + \Delta F_i$. MRT relaxes the non-hydrodynamic moments separately, which keeps the scheme stable at the low viscosity used here.

2 PTFE wall: slip and wetting

NORMAL EQUATIONS · BOUNDARY CONDITIONS ON THE PTFE SURFACE

$$u_t = l_s \frac{\partial u_t}{\partial n} \qquad \text{(Navier slip, as in the paper)}$$

$\theta_r \le \theta \le \theta_a$: contact line carried by the flow; $\theta = \theta_a$ when it advances, $\theta =$

θ_a = 105° (the paper's static angle), θ_r = 20°. Inside this window the wall puts no capillary force on the contact line, which mimics the slowly relaxing wetted-wall condition of the paper's phase-field model.

discretised as

LATTICE BOLTZMANN · BOUNCE-BACK / SPECULAR MIX AND GEOMETRIC WETTING

k_stream9 · _geometric_wall

$$f_i(\mathbf{x}) = r\, f_{\bar{i}}^{*}(\mathbf{x}) + (1-r)\, f_{i'}^{*}\big(\mathbf{x} - e_{ix}\,\hat{\mathbf{x}}\big), \qquad r = 0.02$$

For the upward diagonal directions leaving the PTFE: $\bar{i}$ is the opposite direction (bounce-back, no slip) and i' the mirrored one (specular reflection, free slip). The straight-up direction and all other walls use full bounce-back.

$$\rho_w = \rho_1 + \tan\!\left(\frac{\pi}{2} - \theta\right) \left|\partial_x \rho_1\right|, \qquad \theta = \min\big(\max(\theta_c, \theta_r),\, \theta_a\big), \qquad \theta_c = \frac{\pi}{2} - \arctan$$

$\psi(\mathbf{x}+\mathbf{e}_i) \;\to\; \psi(\rho_w)$ for PTFE neighbours in the Shan–Chen force

ρ_1 and ρ_2 are the densities in the first and second fluid rows above the PTFE, θ_c is the current local contact angle. The plate's end face and the box walls have ψ = 0 (non-wetting).

3 Ions in the droplet

NORMAL EQUATION · NERNST–PLANCK

$$\frac{\partial c}{\partial t} + \nabla\cdot(c\,\mathbf{u}) = \nabla\cdot(D\,\nabla c) \;-\; \nabla\cdot\left(\frac{z\,e\,D}{k_B T}\,c\,\mathbf{E}\right) \qquad \text{(paper, eq. 3)}$$

$$\frac{\partial c}{\partial t} + \nabla\cdot(c\,\mathbf{u}) = \nabla\cdot\left[D\,m\,\nabla\left(\frac{c}{m}\right)\right] \;-\; \nabla\cdot(c\,m\,\mu\,\mathbf{E}) \qquad \text{(this code)}$$

One positive ion species. The diffusion term written with the water fraction m keeps the ions inside the water (equilibrium c ∝ m), which is the paper's no-flux condition at the droplet surface.

discretised as

LATTICE BOLTZMANN · D2Q5 SINGLE-RELAXATION (BGK) ADVECTION–DIFFUSION

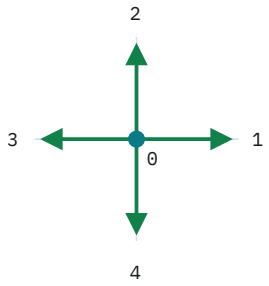
_ion_step · k_ion_collide · k_ion_stream

$$g_i(\mathbf{x}+\mathbf{e}_i,\;t+1) = g_i(\mathbf{x},t) \;-\; \frac{1}{\tau_c}\left[g_i(\mathbf{x},t) - g_i^{eq}(\mathbf{x},t)\right] \qquad i = 0,\dots,4$$

$$g_i^{eq} = w_i\,c\,(1 + 3\,\mathbf{e}_i\cdot\mathbf{v}), \qquad c = \sum_i g_i, \qquad D = \tfrac{1}{3}\left(\tau_c - \tfrac{1}{2}\right)$$

$$\mathbf{v} = \mathbf{u} \;+\; m\,\mu\,\mathbf{E} \;+\; K(m)\,\frac{\nabla m}{\max(m,\,0.02)}, \qquad K(m) = D + k_b\,b(m)$$

τ_c = 0.56 (D = 0.02) · μ = 0.02 · k_b = 0.06 · b(m) = 1 in the interface band (m



D2Q5: $w_0 = 1/3$, $w_{1\dots4} = 1/6$

The ions are carried by the flow $\mathbf{u}$, pulled toward the charged PTFE by $m\,\mu\,\mathbf{E}$, and held inside the water by the last term of $\mathbf{v}$. After every step, negative populations are set to zero and the rest rescaled, so the total charge stays exact.

4 Electric field

NORMAL EQUATION · POISSON (THE PAPER'S GAUSS LAW, EQ. 4.1)

$$\nabla\cdot(\varepsilon\,\nabla\varphi) = -\rho_q, \qquad \rho_q = \sigma_{PTFE}\,\delta_{PTFE} + q\,c, \qquad \mathbf{E} = -\nabla\varphi$$

φ = 0 on the copper film (grounded perfect conductor) and on a grounded boundary 30 mm away. ε = 2 in the plate, 1 in air and in water (a modelling choice; real water is about 80).

discretised as (finite differences, not lattice Boltzmann)

5-POINT FINITE DIFFERENCES · DIRECT SPARSE LU SOLVE EVERY 50 STEPS

solve_poisson · poisson

$$\sum_{nb \in \{E,W,N,S\}} \varepsilon_f \left(\varphi_{nb} - \varphi_P\right) = -\rho_{q,P}, \qquad \varepsilon_f = \frac{2\,\varepsilon_P\,\varepsilon_{nb}}{\varepsilon_P + \varepsilon_{nb}}$$

$$E_x = -\frac{\varphi_{i+1,j} - \varphi_{i-1,j}}{2}, \qquad E_y = -\frac{\varphi_{i,j+1} - \varphi_{i,j-1}}{2}$$

The harmonic mean ε_f keeps the electric displacement continuous across the PTFE/air interface. The matrix is factorised once (ε does not change in time); each later solve takes about 0.2 s.

5 Electrode charge and current

NORMAL EQUATIONS · PAPER EQ. 4.6 AND 4.7

$$\sigma_{Cu} = \mathbf{n} \cdot \mathbf{D}_{PTFE}, \qquad I = \frac{d}{dt} \int \sigma_{Cu}\, dS$$

discretised as

GAUSS'S LAW ON THE COPPER FILM electrode_charge · plot_fig3bc.py

$$Q = -\,\varepsilon_{PTFE} \sum_{i \in Cu} \frac{\varphi_{i,\,PTFE} - \varphi_{i,\,Cu}}{\Delta y} = -\,\varepsilon_{PTFE} \sum_{i \in Cu} \varphi_{i,\,PTFE}, \qquad Q_{nC} = 243\,\frac{Q}{Q_{ref}}$$

The copper film sits one node below the PTFE row at φ = 0. Q_ref is the charge with the PTFE alone (no droplet), so 243 nC means "no droplet". The current is a numerical time derivative, in nA.

6 Lattice and physical units

QUANTITY	LATTICE	PHYSICAL	HOW IT IS MATCHED
Grid spacing Δx	1 (R = 20 nodes)	0.1 mm (R = 2 mm)	droplet radius, paper Table S1

Time step Δt	1	$\approx 20\ \mu s$	capillary time $\sqrt{(\rho R^3/\sigma)}$
Velocity unit $\Delta x/\Delta t$	1	5.0 m/s	Weber number $\rho u^2(2R)/\sigma$
Gravity g	3.9×10^{-5}	9.81 m/s²	Bond number $\rho g R^2/\sigma = 0.545$
Kinematic viscosity v	0.01	5.0×10^{-6} m²/s	5 × water (stability limit)
Surface tension σ	0.056	0.072 N/m	Laplace test of a free droplet